

• Conversion from octal or hexadecimal to binary is done by a procedure reverse to the above. Each octal digit is converted to its three-digit binary equivalent. Similarly, each hexadecimal digit is converted to its four-digit binary equivalent.

$$(673.124)_8 = (\underbrace{110}_6 \underbrace{111}_7 \underbrace{011}_3 \underbrace{001}_1 \underbrace{010}_2 \underbrace{100}_4)_2$$

$$(906.D)_{16} = (\underbrace{1001}_9 \underbrace{0000}_0 \underbrace{0010}_6 \underbrace{1101}_D)_2$$

• During communication between people (about binary numbers in the computer), the octal or hexadecimal representation is more desirable because it can be expressed more compactly with a third or a quarter of the no. of digits required for the equivalent binary number.

• 1.5 Complements

• Complements are used in digital computers for simplifying the subtraction operation and for logical manipulations. There are two types of complements for each base- r system: (1) the r 's complement and (2) the $(r-1)$'s complement.

• When the value of the base is substituted, the two types receive the names 2's and 1's complement for binary numbers, or 10's and 9's complement for decimal numbers.

1.5.1 The r 's Complement

• Given a positive number N in base r with an integer part of n digits, the r 's complement of N is defined as $r^n - N$ for $N \neq 0$ and 0 for $N = 0$.

• Example

• The 10's complement of $(52520)_{10}$ is $10^5 - 52520 = 47480$

• The number of digits in the number is $n=5$.

• The 10's complement of $(0.3267)_{10}$ is $1 - 0.3267 = 0.6733$

• No integer part, so $10^n = 10^0 = 1$.

• The 10's complement of $(25.639)_{10}$ is $10^2 - 25.639 = 74.361$

• The 2's complement of $(101100)_2$ is $(2^6)_{10} - (101100)_2 = (1000000 - 101100)_2$

= 010100

• The 2's complement of $(0.0110)_2$ is $(1 - 0.0110)_2 = 0.1010$

• The 16's complement of $(4BT)_{16}$ is $(16^3 - 4BT)_{10} = (1000 - 4BT)_{16} = (B49)_{16}$